

Santiago Oroz

ARTIFICIAL INTELLIGENCE ENGINEER

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Open to remote work and relocation

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AI Engineer and Higher Technician in Data Science and Artificial Intelligence. I build assistants on **private LLMs (RAG)**, process automation, predictive models and computer vision that **run in production**. Specialised in on-premise deployments where data cannot leave the client's network. **National first place** in the Samsung Innovation Campus 2025 Capstone Project, among more than 30 teams. I own the full cycle: client discovery, development, Linux deployment and ongoing support.

EXPERIENCE

Artificial Intelligence Engineer | Tech del Fuego | Apr 2026 – Present | Río Grande, hybrid

- Member of the **founding team** of a cybersecurity, AI and data science firm serving companies across the region.
- Automation of invoicing, document management and reporting processes, removing hours of manual data entry per week.
- Deployment of **private LLMs** and voice and text WhatsApp assistants wired into CRMs, keeping client data private.
- **Power BI** dashboards unifying scattered databases for predictive analysis and profitability control.
- Data bias auditing and model security review against prompt injection and information leakage.

AI Developer, professional placement | El Dorado SRL | Oct – Dec 2025 | Remote

- Designed and built an **on-premise RAG chatbot** for technical queries on concrete and ISO standards, cutting client response times.
- **Llama 3 running locally** with a **ChromaDB** vector store indexing technical PDFs and an internal MediaWiki: zero data leaving the company network.
- **Flask** API and **React** frontend, with a document administration panel and sources cited in every answer.
- Voice pipeline: speech-to-text with Whisper and text-to-speech with a synthesised Argentine voice.

Data Analysis Intern | Newsan, Motorola plant | Oct – Dec 2024 | Río Grande

- **Business Intelligence** tool letting Health & Safety identify critical workstations and simulate staff rotation schemes to minimise physical fatigue.
- Automation scripts in **VBA and Excel macros** to extract and clean production reports, saving hours of manual work each week.
- **Power BI** dashboards with ergonomic risk heat maps and criticality indices, presented to management and used to rotate staff and prevent injuries.
- Involved across the full data lifecycle: cleaning, normalisation, modelling and visualisation.

PROJECTS

AIDA | Winner, 1st place — Samsung Innovation Campus Capstone | 2025

- **Multimodal** virtual assistant on Telegram for the digital inclusion of older adults. Role: technical co-lead and full stack AI developer, in a team of three.
- **Computer vision** to read paper invoices and detect scam patterns; real-time sentiment analysis that adapts tone and raises risk alerts.
- Modular object-oriented Python architecture, deployed on a **Linux VPS with systemd** to run 24/7 without downtime.
- First place among more than 30 teams and 600 applicants, before a jury from Samsung, Randstad and Streambe. Site: aida-chatbot.github.io/WebAIDA

AXIS, autonomous agents | IBM Hackathon, watsonx | 2026

- Ecosystem of **specialised micro-agents** (Reader, Judge, Analyst, Coordinator and Executor) on Llama-3.2-90b-vision-instruct, connecting legal contracts to daily operations.
- **ReAct** logic and a **mandatory evidence** protocol: no decision is issued without citing the legal document backing it, preventing hallucinations in financial decisions.
- watsonx Orchestrate workflows integrating Jira, Slack, Trello and Microsoft Calendar.

OptiVision | Fuegian Iathon, Fuegian Innovation Agency | 2026

- **Computer vision** platform for industrial ergonomics and quality control on the factory floor, without stopping production. **Working MVP in 8 hours.**
- Webcam pose estimation to detect risky postures, hand gesture detection for contactless operation, a live supervision dashboard and a chatbot over its own data.

Hotel demand forecasting | Machine Learning | 2025

- **XGBoost** regression model forecasting hotel occupancy in Tierra del Fuego, benchmarked against Random Forest as a baseline.
- Exploratory analysis, advanced imputation of missing values and feature engineering on seasonality and calendar variables.

Print Intelligence, expert system | Python | 2025

- **Forward-chaining inference engine** written from scratch, with a JSON knowledge base and step-by-step explanation logic: full traceability, no black boxes.

TECHNICAL SKILLS

Generative AI and LLMs: RAG (Retrieval-Augmented Generation), LangChain, ChromaDB, Llama 3, Llama 4 Scout, LM Studio, Groq API, IBM watsonx, multi-agent architectures, ReAct, prompt engineering, on-premise deployment.

Voice, vision and NLP: Whisper (speech-to-text), Edge-TTS, OCR, pose estimation, gesture detection, pysentimiento, embeddings, vector databases.

Data and machine learning: Python, Pandas, NumPy, Scikit-learn, XGBoost, Random Forest, SQL, Power BI, DAX, Matplotlib, Seaborn, ETL, feature engineering.

Software development: Flask, React, Node.js, Express, MongoDB, JavaScript ES6, Jinja2, object-oriented programming, MVC, Firestore, VBA.

Infrastructure and DevOps: Linux, Bash, VPS management, systemd services, Docker, Git, GitHub, Make and webhook integration.

Security and compliance: NIST CSF 2.0, ISO/IEC 27001, MITRE ATT&CK, algorithmic bias auditing, prompt injection and data leakage prevention.

Languages: Spanish native. English intermediate B1, Infy Institute, one full year of in-person study.

EDUCATION

Full Stack Developer Scholarship, MERN stack — ITBA Executive Education and Santander Open Academy | 2026 | One of 400 selected, 17 intensive weeks

University Technical Degree in Programming — Universidad Tecnológica Nacional (UTN) | 2026 – In progress

Higher Technician in Data Science and Artificial Intelligence — Centro Politécnico Superior Malvinas Argentinas | 2024 – 2025 | Distinguished average, no failed finals

Samsung Innovation Campus, Big Data & AI — Samsung Electronics Argentina | 2025 | 140 hours plus Capstone Project

Artificial Intelligence for Business — Digital House | 2024 – 2025

Secondary degree in Economics and Administration — Instituto Salesiano Don Bosco | 2018 – 2023 | GPA 8.55 / 10